

Discussion 2: The Three Fundamental Effects – Solutions

Names:

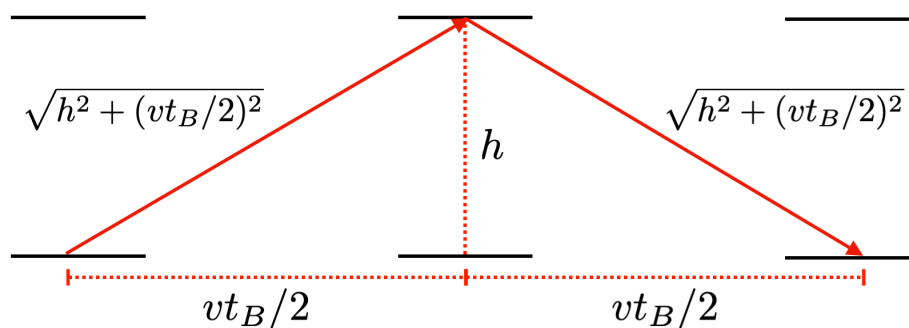
Section:

This week we will get plenty of practice understanding the three fundamental effects of relativity: time dilation, length contraction, and the loss of simultaneity (“rear clock ahead”). We will also see experimental confirmations of the time dilation effect from an experiment using atomic clocks on airplanes.

Part 1: Time Dilation from Light Clock

First let’s derive the time dilation factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ using the thought experiment of a light clock. (Remember that $\gamma \geq 1$ always, because $|v| < c$.) Imagine a train car with mirrors on the floor and ceiling, separated by height h . Person A on the train places a laser pointer on the floor, shines it vertically towards the ceiling, and counts the number of times the light bounces between the mirrors. One “tick” of the light clock is the time t_A it takes the light to make one round trip, which in the frame of the train is $2h/c$. You are on the train tracks watching the train go by with velocity v .

- 1.1. Draw a picture of the path of the laser pointer light in *your* reference frame as it makes one round trip.



- 1.2. Use the Pythagorean Theorem and the constancy of the speed of light to argue that the round-trip time in your frame, t_B , is $\gamma(2h/c)$. Thus $t_B = \gamma t_A$; if 1 second passes in the frame of person A, $\gamma > 1$ seconds pass for you. The moving person experiences less time passing: *moving clocks run slow*.

Light must travel the hypotenuse of the triangle at speed c during a time $t_B/2$, so we must solve

$$\frac{\sqrt{h^2 + (vt_B/2)^2}}{t_B/2} = c.$$

Squaring both sides gives $h^2 + v^2(t_B/2)^2 = c^2(t_B/2)^2$, so $\frac{t_B}{2} = \frac{h}{\sqrt{c^2 - v^2}}$. Rearranging gives $t_B = \frac{2h}{c} \times \gamma$.

- 1.3. A curious and counterintuitive consequence of time dilation occurred to Einstein in 1895, 10 years before he developed the full theory of special relativity: “If one runs after a light wave with [a velocity equal to the] light velocity, then one would encounter a time-independent wavefield.”¹ In other words, *time would stand still* for something traveling at the speed of light! Explain this phenomenon with the light clock by computing the vertical component of the light velocity as a function of v , and taking the limit $v \rightarrow c$.

The ratio of horizontal to vertical components of light velocity must be the ratio of horizontal to vertical distances, so $v_{\text{vert}} = \frac{h}{vt_B/2}c = \frac{2h}{vt_B}c = \frac{c}{\gamma}$. As $v \rightarrow c$, $\gamma \rightarrow \infty$, so the vertical speed goes to zero! If there is no vertical speed, then the light clock can never tick, and time does not elapse.

Part 2: Rear Clock Ahead

The need to synchronize clocks arose as train travel became common across Europe and trains achieved sufficiently high speeds; indeed, it’s possible Einstein’s insights about relativity came from thinking about (and on) trains during his time in a Swiss patent office. Here, we will investigate how weird the world would look if the speed of light were much slower than it actually is! **For the purposes of this problem, pretend that the speed of light is 300 km/hr.**

Imagine this scenario: you are on a train traveling north past Champaign to Chicago (a distance of $L = 210$ km) at velocity $v = 180$ km/hr. When your train passes Champaign, it synchronizes its clock with the station clock so that both read 12:00 noon. The Champaign station clock is also assumed to be synchronized with the Chicago station clock since both are in the same inertial frame.

- 2.1. In the station frame, what time does your train arrive in Chicago? (In other words, what time do the Champaign and Chicago clocks, which are synchronized in the station frame, show when the train arrives in Chicago?)

The train covers the proper distance L in a time $t_{\text{station}} = L/v = 70$ minutes, so the Chicago station clock reads 1:10pm.

- 2.2. What time does *your* clock on the train read, when you arrive in Chicago?

Your time is dilated, so $t_{\text{train}} = L/(\gamma v)$ passes on the train (or equivalently, the distance to Chicago is length-contracted in your frame by the same factor). With the fictitiously slow speed of light, your gamma factor is $\gamma = 5/4$, so $t_{\text{train}} = (4/5)(70 \text{ min}) = 56$ minutes, and your train clock reads 12:56pm.

- 2.3. In your frame on the train, the station clocks are moving. Use the “rear clock ahead” effect to determine the readings on both the Champaign and Chicago clocks in your frame at the moment your train passes Champaign.

¹Translation by Abraham Pais in *Subtle is the Lord*, Oxford University Press, 1982. This is an amazing scientific biography of Einstein which you will enjoy even more after finishing this course!

The Champaign clock reads 12:00 noon in your frame because you synchronized your clock with it. However, the Chicago clock is the “rear” clock in your frame (since Chicago is moving towards you at speed v), and its proper distance is L , so in your frame the Chicago clock has a head start of $Lv/c^2 = 25.2$ minutes. You therefore see the Chicago clock reading 12:25:12.

- 2.4. In your frame, you also see the station clocks running slow because they are moving with respect to you. Combine your results from 4.2 and 4.3 to show that you agree with the reading on the Chicago station clock when you arrive at Chicago. This is consistent with the fact that “train arrives at the Chicago station” is a spacetime event whose coordinates all observers must agree on.

In your frame, it takes 56 minutes to travel to Chicago, but you see the Chicago clock running slowly by a factor $\gamma = 5/4$, and thus $\Delta t_{\text{Chicago}} = (4/5)(56 \text{ min}) = 44.8 \text{ min}$. Combining this with the head start of 25.2 minutes, the Chicago clock reads $25.2 + 44.8 = 70$ minutes past noon in your frame when you arrive in Chicago, which matches the 1:10pm reading as seen by an observer at the station.

Part 3: Clocks on Airplanes

In the *Hafele-Keating experiment*, performed in 1971, ultra-precise atomic clocks were placed on airplanes, one flying westward (opposite the direction of rotation of the Earth) and one flying eastward (with the Earth’s rotation) at the same speed of $v_A = 900 \text{ km/hr}$. The resulting time shifts matched the predictions of special relativity to 10%.

- 3.1. When the airplanes landed and the clocks were compared, the westward clock was found to not only read differently from the eastward clock, it actually *gained* time compared to a clock on the ground! Explain why this is not in conflict with special relativity. (Hint: is a clock on the ground actually in an inertial frame?)

The Earth is rotating at constant velocity, which is not an inertial frame. An eastbound clock will travel faster (and their clock will tick slower) than an observer on the ground, because their speed adds with the instantaneous linear velocity of the Earth. By contrast, a westbound clock will travel slower, and their clock will tick faster.

- 3.2. To calculate the time shifts of the airborne clocks, we need to compare to an inertial frame. Imagine a clock sitting at the center of mass of the Earth, which shows time t_{CM} . Use the rotational velocity of the earth to compute the time dilation factor between t_{CM} and that of a clock sitting on Earth’s surface, t_S . You will find the following approximate expressions helpful:

$$\frac{1}{\sqrt{1 - v^2/c^2}} \approx 1 + \frac{v^2}{2c^2}, \quad \sqrt{1 - v^2/c^2} \approx 1 - \frac{v^2}{2c^2}.$$

Remember that the Earth rotates once in 24 hours, and the radius of the Earth is $R_E \approx 6400 \text{ km}$.²

²You might worry that this computation is invalid because the Earth is a rotating frame, and thus there is a radial acceleration. However, the radial acceleration is in the same direction as the gravitational force, and general relativity tells us that gravity and acceleration are equivalent, so the net effect is very simple: all stationary clocks at the Earth’s surface run at the same rate! That said, there are actually extra time shifts from the effects of general relativity, which predicts that clocks at different heights in a gravitational field run at different rates. Unfortunately we don’t have time to cover this topic in PHYS 225, but feel free to peruse Ch. 5 of Morin.

The Earth's linear velocity is $v_E = (2\pi R_E)/(24 \text{ hr}) = 0.47 \text{ km/s}$. The time dilation factor is approximately $\gamma \approx 1 + \frac{v_E^2}{2c^2} = 1 + 1.2 \times 10^{-12}$. So if $t_{CM} = 1 \text{ second}$, $t_S = t_{CM}/\gamma \approx t_{CM}(1 - 1.2 \times 10^{-12})$ runs slowly by 1.2 picoseconds.

- 3.3. Compute the time dilation of the eastward clock, t_{east} , compared to t_{CM} . By subtracting t_S from t_{east} , calculate how much time (in nanoseconds) the eastward clock lost relative to one on the ground, during the time it took to circle the Earth nonstop.

The velocity of the eastward clock compared to the CM is $v_A + v_E$, so the time dilation factor is

$$t_{east} = t_{CM}/\gamma \approx t_{CM} \left(1 - \frac{(v_A + v_E)^2}{2c^2} \right).$$

Note that $v_A \approx 0.25 \text{ km/s}$, so both terms in the numerator are approximately the same size. Since $t_S \approx t_{CM} \left(1 - \frac{v_E^2}{2c^2} \right)$, we have

$$t_{east} - t_S \approx -t_{CM} \left(\frac{v_A^2 + 2v_E v_A}{2c^2} \right) \approx -1.65 \times 10^{-12} t_{CM}.$$

The absolute difference between t_S and t_{CM} is so small that we can use the time elapsed on the surface, t_S , to approximate the time it takes to circle the Earth: $t_S = \frac{2\pi R_E}{v_A} = 1.6 \times 10^5 \text{ s}$, so $t_{east} - t_S \approx -(1.6 \times 10^5 \text{ s})(1.65 \times 10^{-12}) = -264 \text{ ns}$.

- 3.4. Use the same analysis to calculate the time the westward clock t_{west} *gained* during one nonstop trip around the Earth.

The only difference is that the velocity of the westward clock is $v_E - v_A$ since the plane is going opposite the rotation direction of the Earth. That changes the relative sign in the cross-term:

$$t_{west} - t_S \approx t_{CM} \left(\frac{-v_A^2 + 2v_E v_A}{2c^2} \right) \approx (1.6 \times 10^5 \text{ s})(9.58 \times 10^{-13}) = +153 \text{ ns}.$$